

STANDARD OPERATING PROCEDURE

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Title: In-Service Inspection and Approval of Pressure Vessels

(SYNTHETIC DOCUMENT - created for system testing. Not a real procedure.)

1. PURPOSE

1.1 This procedure defines the requirements for periodic in-service inspection of static pressure vessels and the preparation of the associated approval note.

2. SCOPE

2.1 Applies to all unfired pressure vessels registered under the plant equipment register operating above 1.0 bar gauge.

3. DEFINITIONS

3.1 Nominal Thickness (t_{nom}): the wall thickness recorded at fabrication.

3.2 Minimum Required Thickness (t_{min}): the calculated thickness below which the vessel may not remain in service at its registered design pressure.

3.3 Corrosion Allowance (CA): the sacrificial thickness provided at design stage.

3.4 Corrosion Rate (CR): the rate of wall loss, expressed in millimetres per year.

4. THICKNESS EVALUATION

4.1 Wall thickness shall be measured by ultrasonic testing at not fewer than four locations per shell course.

4.2 The corrosion rate shall be computed as:

$$CR = (t_{previous} - t_{current}) / (\text{years between surveys})$$

4.3 The remaining life shall be computed as:

$$\text{Remaining Life} = (t_{current} - t_{min}) / CR$$

4.4 A vessel with a computed remaining life below 5.0 years shall be referred to the Engineering Review Committee before continued operation is approved.

4.5 A vessel with a computed remaining life below 2.0 years shall be withdrawn from service at the next available opportunity.

5. PRESSURE EVALUATION

5.1 The recorded operating pressure shall be compared against the registered

design pressure. The operating margin shall be computed as:

$$\text{Margin} = \text{Design Pressure} - \text{Operating Pressure}$$

5.2 An operating margin below 2.0 bar shall be recorded as a non-conformance and escalated to the Head of Mechanical Inspection.

5.3 Operation above the registered design pressure is prohibited without a formal de-rating or re-rating study approved under SOP-MECH-021.

6. APPROVAL NOTE REQUIREMENTS

6.1 An approval note prepared under this procedure shall contain, as a minimum:

- (a) equipment tag and description;
- (b) inspection reference number and date;
- (c) each measured value, with its source document, page and location;
- (d) each derived value, with the formula and inputs used;
- (e) an explicit statement of compliance or non-compliance against clause 4.4, clause 4.5 and clause 5.2;
- (f) the recommendation and the name of the recommending engineer;
- (g) a signature block for the approving authority.

6.2 Any value which cannot be established from the inspection record shall be recorded as "CANNOT DETERMINE" together with the information required to resolve it. Estimated or assumed values shall not be entered into an approval note.

7. RETENTION

7.1 Inspection records and approval notes shall be retained for the operating life of the equipment plus ten years.

END OF PROCEDURE